

Polynomial divisibility of height-one factorial ratios

Proofs of Bala’s uniform-multiplier conjectures,
a complete rational-root criterion,
and certified optimal constants

A research attempt with reproducible exact certificates
September 19, 2026

Main outcome

For

$$A(n) = \frac{(30n)! n!}{(15n)! (10n)! (6n)!},$$

we prove the all- r divisibility conjectures still recorded under OEIS A211417. We also classify every polynomial splitting over $\mathbb{Q}$ that can uniformly divide $A(n)$ after multiplication by one fixed integer. The analogous conjectures for OEIS A295431 follow from the same theorem.

Abstract. Let $R(n)$ be a balanced integral factorial ratio and let $P \in \mathbb{Z}[X]$ split over $\mathbb{Q}$. We characterize the existence of a nonzero integer C with $CR(n)/P(n) \in \mathbb{Z}$ for every nonnegative integer n at which the quotient is defined. The criterion compares the multiplicity of each rational root of P with one-sided values of the Landau step function. We prove both directions and give an explicit multiplier. At height one, the criterion reduces to the signs of cyclotomic exponents and excludes all repeated roots. Applications settle uniform-product conjectures of Peter Bala, including explicit all- r bounds. Finite weighted digit graphs determine the eight particular proposed constants and prove each optimal. The associated integer sequences admit explicit recurrences, hypergeometric generating functions, and all-order asymptotic expansions. The archive contains exact certificates, a separate verifier, regression tests, and an admissible-denominator atlas for all 52 sporadic examples.

Contents

1	The selected problem and its status	2
1.1	Two exceptional integer sequences	2
1.2	Results obtained	2
1.3	What is and is not asserted about novelty	3
2	Factorial valuations and the Landau function	3
2.1	Definitions and a self-contained integrality criterion	3
2.2	Reflection and jumps	4
3	A complete criterion for polynomials with rational roots	5
3.1	The one-sided capacity	5
3.2	Proof of sufficiency	5
3.3	Proof of necessity	6
3.4	Consequences and computational meaning	7
4	The height-one simplification	7
4.1	Cyclotomic interpretation	7
5	Resolution of the uniform-product conjectures	8
5.1	The two exact slope lists	8
5.2	Sharper explicit bounds for the prescribed products	8
5.3	Removing a restriction on positive offsets	9
5.4	An explicit infinite obstruction family	9
6	Exact optimal multipliers by finite certificates	10
6.1	Why checking a large interval is not enough	10
6.2	The state and transition rules	10
6.3	Potential certificates	11
6.4	Results of the exact verification	11
7	Integer sequences, recurrences, and generating functions	12
7.1	An integer sequence singled out by the optimal constant	12
7.2	A hypergeometric representation	13
7.3	Differential and integral transforms	13
8	All-order asymptotic expansions	14
8.1	The general factorial-ratio expansion	14
8.2	The expansion for the certified sequence	15
9	Verification record, limitations, and next questions	15
9.1	What was actually checked	15
9.2	Limits of the conclusions	16
9.3	Research directions arising from the proof	16
A	Admissible denominators for all 52 sporadic ratios	16
B	Reproducing the certificates and data	18

1 The selected problem and its status

1.1 Two exceptional integer sequences

Consider the factorial ratios

$$A(n) = \frac{(30n)!n!}{(15n)!(10n)!(6n)!}, \quad B(n) = \frac{(12n)!n!}{(6n)!(4n)!(3n)!}, \quad n \geq 0. \quad (1)$$

They are OEIS A211417 and A295431, respectively. Both belong to the 52 sporadic height-one integral factorial ratios in the classification of Bober, following the work of Vasyunin and the hypergeometric approach of Rodriguez-Villegas and Beukers–Heckman [1, 2, 4]. Our arguments require no classification of hypergeometric monodromy.

Bala’s August 2025 comments on A211417 ask for a fixed multiplier, depending on k and r but not on n , that clears the denominators in

$$\frac{A(n)}{\prod_{\substack{1 \leq i \leq r \\ (i,k)=1}} (kn+i)}, \quad k \in \{2, 3, 5\}. \quad (2)$$

There is also the negative-offset family

$$\frac{A(n)}{\prod_{\substack{1 \leq i \leq r \\ (i,30)=1}} (30n-i)}. \quad (3)$$

The current entry still labels these all- r assertions conjectural [1]. Its one-factor case $A(n)/(30n-1)$ is different: the entry records a proof in June 2026, and Adamczewski’s August 2026 report also lists it as proved [1, 5]. We use it only as a previously known special case recovered by a broader argument.

The companion A295431 entry poses similar uniform-product questions for $B(n)$, using positive slopes 1, 2, 3 and negative factors $12n-i$ with $(i, 12) = 1$ [2]. The selected problem is therefore not merely to reprove integrality of (1), or to reproduce the solved first negative-offset case. It is to explain and prove the uniform-product phenomenon.

1.2 Results obtained

Write

$$\mathcal{L}_s = \text{lcm}(1, 2, \dots, s), \quad \mathcal{L}_1 = 1, \quad m_k(r) = \#\{1 \leq i \leq r : (i, k) = 1\}.$$

Our first concrete conclusions are

$$\mathcal{L}_{30r/k}^{m_k(r)} \frac{A(n)}{\prod_{1 \leq i \leq r, (i,k)=1} (kn+i)} \in \mathbb{Z} \quad (k = 2, 3, 5), \quad (4)$$

$$\mathcal{L}_r^{m_{30}(r)} \frac{A(n)}{\prod_{1 \leq i \leq r, (i,30)=1} (30n-i)} \in \mathbb{Z}. \quad (5)$$

These hold for every $r \geq 1$ and $n \geq 0$. The indices $30r/k$ are integers. The displayed constants are deliberately explicit rather than claimed to be smallest. The corresponding negative-offset formula for B uses $\mathcal{L}_r^{m_{12}(r)}$.

The structural result is stronger. A polynomial splitting over $\mathbb{Q}$ can divide $A(n)$ uniformly up to one fixed integer multiplier precisely when it has distinct, nonzero rational roots, and the reduced denominator of each negative root lies in $\{1, 2, 3, 5\}$, while the reduced denominator of each positive root is 30. For B , these sets become $\{1, 2, 3\}$ and $\{12\}$. This also proves that the positive-offset coprimality restrictions can be removed, at the cost of changing the constant.

The eight particular constants in the two OEIS entries are correct and optimal:

Ratio	Dividing polynomial	Least positive multiplier
A	$2n + 1$	7
A	$3n + 1$	1
A	$5n + 1$	1
A	$(2n + 1)(3n + 1)(5n + 1)$	42
B	$n + 1$	385
B	$2n + 1$	5
B	$3n + 1$	1
B	$(n + 1)(2n + 1)(3n + 1)$	770

Existence for these eight cases follows already from the structural proof. Their exact constants are established by the finite certificates in Section 6.

1.3 What is and is not asserted about novelty

The source check was made on September 19, 2026. It distinguishes the still-conjectural all- r statements from the explicitly solved $30n - 1$ case. The machinery of factorial valuations and Landau step functions is classical. We give a complete proof of the additional polynomial-divisibility statements here, but do not claim that an exhaustive literature search has ruled out every equivalent formulation of the general criterion. In particular, this manuscript's mathematical claims are separable from a claim of publication priority.

2 Factorial valuations and the Landau function

2.1 Definitions and a self-contained integrality criterion

Let $a_1, \dots, a_u$ and $b_1, \dots, b_v$ be positive integers satisfying

$$\sum_{i=1}^u a_i = \sum_{j=1}^v b_j. \quad (6)$$

Set

$$R(n) = \frac{\prod_{i=1}^u (a_i n)!}{\prod_{j=1}^v (b_j n)!}, \quad \Delta(x) = \sum_{i=1}^u [a_i x] - \sum_{j=1}^v [b_j x], \quad h = v - u. \quad (7)$$

The integer h is the height. The balance condition makes Δ periodic with period one, since the terms linear in an integer translation cancel. It is a right-continuous integer-valued step function. Put $M = \max(a_i, b_j)$.

For a prime p , $v_p(q)$ denotes its exponent in a nonzero rational number q ; it may be negative. Counting the multiples of $p, p^2, \dots$ in a factorial gives

$$v_p(N!) = \sum_{e \geq 1} \lfloor N/p^e \rfloor, \quad v_p(R(n)) = \sum_{e \geq 1} \Delta(n/p^e). \quad (8)$$

All these sums are finite. A nonzero rational number is an integer if and only if all its prime valuations are nonnegative.

Lemma 2.1 (One-variable Landau criterion). *Under (6), $R(n) \in \mathbb{Z}$ for every $n \geq 0$ if and only if $\Delta(x) \geq 0$ for every real x .*

Proof. Nonnegativity gives integrality immediately from (8). Conversely, suppose Δ is negative somewhere. Right continuity and the finite set of discontinuities in a period give a nonempty open interval $(\alpha, \beta) \subset (0, 1)$ on which it is negative. Choose an arbitrarily large prime p and an integer n with $\alpha < n/p < \beta$; such an n exists once $p(\beta - \alpha) > 1$. For $p > M$ we have $0 \leq n/p^e < 1/M$ for $e \geq 2$. On this last interval every floor in (7) is zero. Thus $v_p(R(n)) = \Delta(n/p) < 0$, contradicting integrality. Periodicity handles every real x . $\square$

This is the relevant specialization of Landau's theorem; Bober gives its general setting and consequences [4, Section 3]. Bala's note supplies a direct proof for A and its floor-factorial extension [3].

2.2 Reflection and jumps

For $k \geq 1$, define

$$J(k) = \#\{i : k \mid a_i\} - \#\{j : k \mid b_j\}. \quad (9)$$

If $(t, k) = 1$, the jump at t/k is

$$\Delta(t/k) - \Delta((t/k)^-) = J(k). \quad (10)$$

Indeed, $\lfloor cx \rfloor$ increases at t/k precisely when $k \mid c$. Here $\Delta(x^-)$ is the value just to the left of x ; the value just to the right is $\Delta(x)$ itself.

Away from the discontinuities,

$$\Delta(x) + \Delta(1 - x) = h. \quad (11)$$

To see this, use $\lfloor cx \rfloor + \lfloor c(1 - x) \rfloor = c - 1$ when cx is not an integer and sum over the parameter lists. At a reduced rational point the endpoint version is

$$\Delta(t/k) + \Delta(1 - t/k) = h + J(k), \quad (12)$$

$$\Delta((t/k)^-) + \Delta((1 - t/k)^-) = h - J(k). \quad (13)$$

These follow either by counting the integral cx separately or by subtracting the two jumps. The formulas remain valid at integers by periodicity.

Corollary 2.2. *For an integral balanced ratio of height one, every value and one-sided value of Δ belongs to $\{0, 1\}$, and $J(k) \in \{-1, 0, 1\}$.*

Proof. At a point avoiding the discontinuities, the two nonnegative integers in (11) have sum one. Right continuity gives the same range at every point, and left limits have the same range. Equation (10) then bounds the jumps. $\square$

3 A complete criterion for polynomials with rational roots

3.1 The one-sided capacity

Every nonzero polynomial in $\mathbb{Z}[X]$ that splits over $\mathbb{Q}$ can be written

$$P(X) = c \prod_{j=1}^s (k_j X + \ell_j)^{m_j}, \quad (14)$$

where $c \in \mathbb{Z} \setminus \{0\}$, $k_j > 0$, $(k_j, \ell_j) = 1$, $m_j \geq 1$, and the roots $-\ell_j/k_j$ are pairwise distinct. This is primitive factorization over $\mathbb{Z}$ after collecting equal roots. Write $d = \sum_j m_j$. A constant polynomial is allowed by taking $s = 0$.

For $\ell \neq 0$, define the capacity

$$\kappa_R(k, \ell) = \begin{cases} \min_{\substack{1 \leq t \leq k \\ (t, k) = 1}} \Delta((t/k)^-), & \ell > 0, \\ \min_{\substack{1 \leq t \leq k \\ (t, k) = 1}} \Delta(t/k), & \ell < 0. \end{cases} \quad (15)$$

For $k = 1$, the only representative is $t = 1$; thus the left capacity is h and the right capacity is zero. We also set the capacity of the primitive zero-root factor X equal to zero.

The sign matters. A positive offset $kn + \ell$ forces n/p^e to approach a reduced rational point from the left. A negative offset forces approach from the right. The size of $|\ell|$ affects the exceptional small prime powers, but not the capacity.

Theorem 3.1 (Polynomial divisibility criterion). *Assume R is balanced and integral. For P as in (14), the following are equivalent:*

- (i) *There is a positive integer C such that $CR(n)/P(n) \in \mathbb{Z}$ for every $n \geq 0$ with $P(n) \neq 0$.*
- (ii) *The polynomial has no zero root and $m_j \leq \kappa_R(k_j, \ell_j)$ for every j .*

When (ii) holds and $s > 0$, one valid choice is

$$C = |c| \mathcal{L}_T^d, \quad T = \max \left(1, M \max_j |\ell_j|, \max_{i < j} |k_i \ell_j - k_j \ell_i| \right), \quad (16)$$

with the last maximum omitted if $s = 1$. For $s = 0$, $C = |c|$ works.

3.2 Proof of sufficiency

We isolate the large-prime-power part of the proof.

Lemma 3.2. *Suppose $\ell \neq 0$, $(k, \ell) = 1$, and $q = p^e > M|\ell|$. If $q \mid kn + \ell$ and $kn + \ell \neq 0$, then*

$$\Delta(n/q) \geq \kappa_R(k, \ell).$$

Proof. If $p \mid k$, then $p \nmid \ell$, so divisibility is impossible. Otherwise put $kn + \ell = tq$. Since $q > |\ell|$ and $n \geq 0$, a nonzero multiple $tq = kn + \ell$ cannot be negative; hence $t \geq 1$. Reducing modulo k gives $(t, k) = 1$, since both q and ℓ are coprime to k . We have

$$\frac{n}{q} = \frac{t}{k} - \frac{\ell}{kq}. \quad (17)$$

Every discontinuity of Δ has the form v/a_i or v/b_j . A discontinuity different from t/k is at distance at least $1/(kM)$: a nonzero difference of these rational numbers has numerator at least one after placing it over denominator ka_i or kb_j . The displacement in (17) is strictly smaller than this. Thus $\Delta(n/q)$ equals the left value at t/k if $\ell > 0$, and the right value if $\ell < 0$. Periodicity reduces t to the representatives in (15). $\square$

Sufficiency in Theorem 3.1. Let $Q(X) = \prod_j (k_j X + \ell_j)^{m_j}$. For a prime power q , put

$$E_q(n) = \sum_{j=1}^s m_j \mathbf{1}_{q|k_j n + \ell_j}.$$

If $q > T$, it cannot divide two distinct primitive factors. Otherwise it would divide the nonzero integer $k_i \ell_j - k_j \ell_i$, whose absolute value is at most T . Consequently, for $q > T$, either $E_q(n) = 0$ or exactly one root contributes, with its multiplicity m_j . Lemma 3.2 and (ii) give

$$\Delta(n/q) - E_q(n) \geq 0 \quad (q > T). \quad (18)$$

For $q \leq T$, Landau nonnegativity and $E_q(n) \leq d$ give the lower bound $-d$. Therefore, for every prime p ,

$$\begin{aligned} v_p(R(n)/Q(n)) &= \sum_{e \geq 1} (\Delta(n/p^e) - E_{p^e}(n)) \\ &\geq -d \# \{e \geq 1 : p^e \leq T\} = -dv_p(\mathcal{L}_T). \end{aligned}$$

This proves $\mathcal{L}_T^d R(n)/Q(n) \in \mathbb{Z}$. Since $P = cQ$, multiplying by $|c|$ only adds the harmless sign $|c|/c$. No unproved assertion about an infinite computational range enters the argument. $\square$

3.3 Proof of necessity

The converse explains why the allowed slopes are restrictive.

Necessity in Theorem 3.1. First suppose P has a zero root. Primitivity makes its factor X . For any sufficiently large prime p , take $n = p$. Then $\Delta(n/p) = \Delta(1) = 0$, and $\Delta(n/p^e) = 0$ for $e \geq 2$. Hence $v_p(R(p)) = 0$, while $p \mid P(p)$, except for finitely many p that could make $P(p) = 0$. No fixed multiplier can supply every such prime.

Now fix a nonzero root with primitive factor $kX + \ell$, multiplicity m , and capacity $\kappa < m$. Choose a reduced representative t/k at which the relevant one-sided value is κ . Dirichlet's theorem provides infinitely many primes in the progression

$$p \equiv \ell t^{-1} \pmod{k}; \quad (19)$$

this is an eligible class because $(t, k) = (\ell, k) = 1$. For $k = 1$, it means all primes. We use only the infinitude assertion of Dirichlet's theorem [6, Theorem 4.2].

For such a sufficiently large prime set

$$n = \frac{tp - \ell}{k}. \quad (20)$$

Then n is a nonnegative integer and $kn + \ell = tp$. Taking $p > k$ ensures $p \nmid t$, so the factor has valuation exactly one, and its m th power contributes m to $v_p(P(n))$. As p tends to infinity,

$n/p = t/k - \ell/(kp)$ approaches from the required side; thus $\Delta(n/p) = \kappa$ eventually. For all sufficiently large p , $Mn < p^2$, so every term with $e \geq 2$ vanishes. It follows that

$$v_p(R(n)) = \kappa < m \leq v_p(P(n)).$$

The values (20) tend to infinity, so finitely many zeros of P do not obstruct the construction. Infinitely many different primes appear in reduced denominators of $R(n)/P(n)$, ruling out every fixed nonzero multiplier. $\square$

3.4 Consequences and computational meaning

The criterion is effective. All relevant values of Δ and all jumps can be computed with exact integer or rational arithmetic. It separates the problem into two very different tasks: deciding whether *some* multiplier exists, and finding the *least* multiplier. The first is answered by a finite root-by-root test. The second must account for compensation between valuation contributions at different powers of the same small prime.

The multiplicity condition is also essential. The fact that a sequence is divisible by one linear expression, up to a fixed constant, does not imply the same for its square. Higher-height ratios can support larger multiplicities: for example $A(n)^2$ supports $(2n+1)^2$, but not $(2n+1)^3$. This follows either from the theorem or from doubling the Landau function.

4 The height-one simplification

Theorem 4.1 (Jump-sign classification). *Suppose R is balanced, integral, and of height one. For a primitive factor $kX + \ell$ with $\ell \neq 0$, its capacity is one exactly when*

$$J(k) = -1 \quad \text{if } \ell > 0, \quad J(k) = +1 \quad \text{if } \ell < 0. \quad (21)$$

Otherwise its capacity is zero. Thus the admissible polynomials are exactly those with distinct nonzero roots whose primitive factors satisfy (21).

Proof. By Corollary 2.2, every value is zero or one. If $J(k) = -1$, then (10) forces the left value to be one and the right value to be zero at *every* reduced t/k . Conversely, if all left values are one, pair t/k with $1 - t/k$ in (13). Their sum is two, so $1 - J(k) = 2$ and $J(k) = -1$. The same argument with (12) proves that all right values are one exactly when $J(k) = +1$. This includes the self-paired points and $k = 1$, so no endpoint exception is needed. All capacities are at most one, and Theorem 3.1 finishes the proof. $\square$

4.1 Cyclotomic interpretation

Let $\Phi_k(z)$ be the k th cyclotomic polynomial. Factorization of $z^m - 1$ gives the rational-function identity

$$\frac{\prod_i (z^{a_i} - 1)}{\prod_j (z^{b_j} - 1)} = \prod_{k \geq 1} \Phi_k(z)^{J(k)}. \quad (22)$$

Only finitely many exponents are nonzero. At height one the allowed positive-offset denominators are exactly the cyclotomic factors remaining downstairs after cancellation in (22); the allowed

negative-offset denominators are those remaining upstairs. This is an interpretation of the proved criterion, not an additional monodromy assumption.

For the binomial example $R(n) = \binom{2n}{n}$, the sets are $\{1\}$ and $\{2\}$. Accordingly, fixed products of distinct $n + i$ can be divided out up to constants, as can fixed products of distinct $2n - i$ with i odd, but a repeated such root is impossible. For the 52 sporadic ratios, Appendix A computes the two sets explicitly from their parameter lists.

5 Resolution of the uniform-product conjectures

5.1 The two exact slope lists

For A , direct cancellation in (22) gives

$$\frac{(z^{30} - 1)(z - 1)}{(z^{15} - 1)(z^{10} - 1)(z^6 - 1)} = \frac{\Phi_{30}(z)}{\Phi_1(z)\Phi_2(z)\Phi_3(z)\Phi_5(z)}. \quad (23)$$

For B ,

$$\frac{(z^{12} - 1)(z - 1)}{(z^6 - 1)(z^4 - 1)(z^3 - 1)} = \frac{\Phi_{12}(z)}{\Phi_1(z)\Phi_2(z)\Phi_3(z)}. \quad (24)$$

These identities prove the slope lists announced in Section 1, provided integrality is known. For completeness, it can be checked particularly simply here. On the grid interval $j/30 \leq x < (j+1)/30$,

$$\Delta_A(x) = j - \lfloor j/2 \rfloor - \lfloor j/3 \rfloor - \lfloor j/5 \rfloor, \quad 0 \leq j < 30.$$

The three subtracted floors have sum at most $\lfloor 31j/30 \rfloor = j$. This proves nonnegativity and hence integrality of A . For B the corresponding formula is

$$\Delta_B(x) = j - \lfloor j/2 \rfloor - \lfloor j/3 \rfloor - \lfloor j/4 \rfloor, \quad 0 \leq j < 12,$$

and the sum is at most $\lfloor 13j/12 \rfloor = j$. Height one then gives the upper bound one in each case.

Corollary 5.1. *Let $P \in \mathbb{Z}[X] \setminus \{0\}$ split over $\mathbb{Q}$. There exists a fixed positive integer C with $CA(n)/P(n) \in \mathbb{Z}$ wherever defined on $\mathbb{Z}_{\geq 0}$ if and only if P has distinct nonzero roots, every negative root in lowest terms has denominator 1, 2, 3, or 5, and every positive root in lowest terms has denominator 30. For B , replace these lists by 1, 2, 3 and 12.*

5.2 Sharper explicit bounds for the prescribed products

The general bound (16) already establishes existence. The grid structure gives the cleaner formulas (4)–(5).

Proposition 5.2. *For $k \in \{2, 3, 5\}$, let $I = \{1 \leq i \leq r : (i, k) = 1\}$ and $m = |I|$. Then the multiplier $\mathcal{L}_{30r/k}^m$ works for $A(n)/\prod_{i \in I}(kn + i)$.*

Proof. If $q = p^e > 30r/k$ divides $kn + i$, then $p \nmid k$ and $n/q = t/k - i/(kq)$ with $(t, k) = 1$. The left grid interval of width $1/30$ at t/k has $\Delta_A = 1$, since $J_A(k) = -1$. The displacement is smaller than $1/30$, so $\Delta_A(n/q) = 1$. Moreover, $q > r$ cannot divide two factors, since their difference is a nonzero integer $i - j$ of absolute value less than r . For smaller q the total deficit is at worst m . The same valuation sum used in Theorem 3.1 gives the proposed multiplier. $\square$

Proposition 5.3. *Let (R, M) be either $(A, 30)$ or $(B, 12)$ and let $I = \{1 \leq i \leq r : (i, M) = 1\}$. Then*

$$\mathcal{L}_r^{|I|} \frac{R(n)}{\prod_{i \in I} (Mn - i)} \in \mathbb{Z} \quad (n \geq 0). \quad (25)$$

Proof. No denominator factor is zero, since i is coprime to $M > 1$. If $q > r$ divides $Mn - i$, then $p \nmid M$ and $n/q = t/M + i/(Mq)$ with $(t, M) = 1$. The right grid interval of width $1/M$ at t/M has value one, because $J_R(M) = +1$. Hence the corresponding Landau term pays for this divisor. Two factors cannot both be divisible by $q > r$, because their difference has absolute value less than r . The low prime powers contribute a deficit of at most $|I|$ each, exactly as before. $\square$

For $r = 1$ the multiplier is one. Thus the proposition recovers the previously proved $A(n)/(30n-1)$ theorem without depending on its proof. The purpose of the proposition is its uniform extension to every r . Negative values at $n = 0$ cause no problem: divisibility and valuations use the absolute value of a nonzero denominator.

5.3 Removing a restriction on positive offsets

For $R = A$ and $k \in \{1, 2, 3, 5\}$, or for $R = B$ and $k \in \{1, 2, 3\}$, consider the unrestricted product

$$P_{k,r}(n) = \prod_{i=1}^r (kn + i).$$

Put $g_i = (k, i)$ and reduce its factors to $(k/g_i)n + i/g_i$. Their roots are still distinct, and their reduced slopes divide k . Every such divisor is in the corresponding positive-offset list. Consequently a uniform multiplier exists. A fully explicit choice, with $M = 30$ or 12 respectively, is

$$\left(\prod_{i=1}^r g_i \right) \mathcal{L}_{Mr}^r. \quad (26)$$

Indeed, the primitive offsets have absolute value at most r , and their pairwise resultants are $k(j-i)/(g_i g_j)$, of absolute value less than $kr \leq Mr$. Theorem 3.1 applies, with the product of the g_i accounting for the content of the polynomial.

In contrast, the negative-offset coprimality restrictions are structural. Reducing $30n - i$ when $(30, i) > 1$ gives a proper divisor of 30 as its slope, none of which is in the negative-offset list for A . A uniform multiplier for that one factor is impossible. The same statement holds for $12n - i$ and B .

5.4 An explicit infinite obstruction family

The slope 4 is not allowed for positive offsets of A . This failure can be exhibited without relying on a table of numerical experiments. Let $p > 30$ be a prime with $p \equiv 3 \pmod{4}$, and set

$$n = \frac{3p-1}{4}.$$

Then $4n+1 = 3p$, whereas $n/p = 3/4 - 1/(4p)$ lies in $[22/30, 23/30)$, where $\Delta_A = 0$. Also $30n < p^2$ for $p > 30$, so all higher valuation terms vanish. Thus $v_p(A(n)) = 0$ while $p \mid 4n+1$. Infinitely many distinct such primes exist, so no constant clears all denominators of $A(n)/(4n+1)$. The first example in this family is $p = 31$, $n = 23$.

6 Exact optimal multipliers by finite certificates

6.1 Why checking a large interval is not enough

The large-prime-power proof gives a constant, but not necessarily the smallest one. It is not valid to maximize the deficit at each exponent independently and claim that the resulting sum is sharp: residues at $p, p^2, p^3, \dots$ are coupled by the base- p digits of n . For instance, an unfavorable contribution at p can force a favorable contribution at p^2 or later. This is why some quotients above need no multiplier even though an individual Landau summand can fail to cover the denominator at a small prime power.

We encode all possible digit continuations in a finite directed graph. Its integer edge weights add to the exact valuation, not to a floating point approximation. A feasible potential on the graph proves a lower bound for every possible n simultaneously.

6.2 The state and transition rules

For this section let $R = A$ or B , let $L = 30$ or 12 respectively, and let

$$Q(n) = \prod_{k \in K} (kn + 1),$$

where K is one of the eight slope sets in the result table. Write $\delta(j) = \Delta_R(j/L)$ for $0 \leq j < L$. Process a base- p expansion of n from least significant digit to most significant digit. After e digits, let $r_e = n \bmod p^e$ and set

$$j_e = \lfloor Lr_e/p^e \rfloor.$$

Because every factorial parameter divides L , the valuation summand $\Delta_R(n/p^e)$ equals $\delta(j_e)$.

A factor $kn + 1$ is called alive after e digits when $p^e \mid kr_e + 1$. For an alive factor retain the integer carry

$$t_{k,e} = (kr_e + 1)/p^e.$$

Its possible values are $1, \dots, k$. Once a factor is not alive, it can never be alive again, since divisibility by p^{e+1} implies divisibility by p^e . Mark it by $\perp$ and forget its carry. The initial state is

$$s_0 = (0, 1, \dots, 1).$$

If the next digit is $d \in \{0, \dots, p-1\}$, the next state satisfies

$$j' = \lfloor (j + Ld)/p \rfloor, \tag{27}$$

and for each factor

$$t'_k = \begin{cases} (t_k + kd)/p, & t_k \neq \perp \text{ and } p \mid t_k + kd, \\ \perp, & \text{otherwise.} \end{cases} \tag{28}$$

To verify (27), write $Lr_e/p^e = j + \varepsilon$ with $0 \leq \varepsilon < 1$; dividing $j + Ld + \varepsilon$ by the integer p does not alter the floor obtained from $j + Ld$. The carry formula follows by replacing r_e with $r_e + dp^e$.

Give this transition the weight

$$w(s, d) = \delta(j') - \#\{k \in K : t'_k \neq \perp\}. \tag{29}$$

Reading all the digits of n and then enough zero digits reaches

$$s_* = (0, \perp, \dots, \perp).$$

Indeed $j_e = 0$ when $p^e > Ln$, and no positive integer $kn + 1$ is divisible by every power of p . The terminal state remains fixed under further zero digits and has zero edge weight. Equations (8) and (29) show that the sum along this finite path is exactly

$$v_p(R(n)/Q(n)). \quad (30)$$

6.3 Potential certificates

Proposition 6.1 (Finite certificate rule). *Let S be a finite collection of states containing s_0 and s_* and closed under every digit transition. Suppose an integer function $H : S \rightarrow \mathbb{Z}$ satisfies*

$$H(s_*) = 0, \quad H(s) \leq w(s, d) + H(s') \quad \text{for every } s \in S \text{ and every digit } d. \quad (31)$$

Then $v_p(R(n)/Q(n)) \geq H(s_0)$ for every $n \geq 0$. If some explicit n_0 has valuation $H(s_0)$, the lower bound is sharp.

Proof. Apply the edge inequality successively along the path for n described above. The intermediate values of H telescope, leaving $H(s_0) \leq \sum w + H(s_*) = v_p(R(n)/Q(n))$. An equality witness proves that no larger lower bound is possible. $\square$

The generator enumerates reachable states and computes shortest costs to the terminal by integer relaxation. This search is *not* trusted as the proof: a separate program reads the resulting state list and potential and checks every closure condition and inequality in (31). It also checks the equality witness directly from Legendre's formula. All prime certificates needed for a particular quotient are checked for complete coverage.

6.4 Results of the exact verification

For A the general cutoff (16) is $T = 30$ for these positive unit offsets; for B it is $T = 12$. Thus primes larger than these bounds need no certificate: each of their valuation summands is nonnegative after subtracting the denominator indicator. Below the cutoff the exact minima are as follows. A zero beyond a row's cutoff is both a lower bound from the theorem and an attained minimum at $n = 0$.

$(R; K)$	2	3	5	7	11	13	17	19	23	29	$C_{\min}$
$(A; 2)$	0	0	0	-1	0	0	0	0	0	0	7
$(A; 3)$	0	0	0	0	0	0	0	0	0	0	1
$(A; 5)$	0	0	0	0	0	0	0	0	0	0	1
$(A; 2, 3, 5)$	-1	-1	0	-1	0	0	0	0	0	0	42
$(B; 1)$	0	0	-1	-1	-1	0	0	0	0	0	385
$(B; 2)$	0	0	-1	0	0	0	0	0	0	0	5
$(B; 3)$	0	0	0	0	0	0	0	0	0	0	1
$(B; 1, 2, 3)$	-1	0	-1	-1	-1	0	0	0	0	0	770

The headings 2, 3, ... are primes, and the entries are $\min_{n \geq 0} v_p(R(n)/Q(n))$. The table is not inferred from testing an interval: it is the output of the checked finite potentials.

The nonzero lower-bound requirements are all necessary, as witnessed by the following small indices:

$(R; K)$	Prime	Index with valuation -1
$(A; 2)$	7	10
$(A; 2, 3, 5)$	2, 3, 7	3, 16, 10, respectively
$(B; 1)$	5, 7, 11	9, 20, 10, respectively
$(B; 2)$	5	2
$(B; 1, 2, 3)$	2, 5, 7, 11	1, 2, 20, 10, respectively

For each prime, any valid multiplier must contain that prime to at least the indicated deficit. Multiplying these independently necessary prime powers gives the least multiplier; the indices need not be the same for different primes. This proves optimality of all eight constants.

The delivered certificates comprise **60 prime graphs, 1,685 states, and 19,277 checked transition inequalities**. Their compact JSON representation contains every state and every potential value, together with equality witnesses. The separate verifier recomputes all transitions instead of trusting stored edges.

7 Integer sequences, recurrences, and generating functions

7.1 An integer sequence singled out by the optimal constant

Define

$$U_n = \frac{42A(n)}{(2n+1)(3n+1)(5n+1)}. \quad (32)$$

The preceding section proves $U_n \in \mathbb{Z}_{>0}$, with the factor 42 minimal for integrality at every $n \geq 0$. Its first terms are

$$\begin{aligned} U_0 &= 42, \\ U_1 &= 45287852610, \\ U_2 &= 5873158887743776788432, \\ U_3 &= 1645528285131615053352729526247883. \end{aligned}$$

The archive supplies exact terms through $n = 50$. We make no assertion that this sequence lacks an existing OEIS entry; the contribution here is its proved uniform integrality and optimal normalization.

For an arbitrary polynomial divisor P covered by our theorem, $u_n = CR(n)/P(n)$ has the explicit first-order recurrence

$$\frac{u_{n+1}}{u_n} = \frac{P(n)}{P(n+1)} \frac{\prod_{i=1}^u \prod_{t=1}^{a_i} (a_i n + t)}{\prod_{j=1}^v \prod_{t=1}^{b_j} (b_j n + t)}. \quad (33)$$

Cross-multiplication gives a recurrence with polynomial coefficients. This follows directly by canceling consecutive factorials and is valid wherever the quotient is defined. In our admissible cases P has no nonnegative integer root, so there is no missing index.

7.2 A hypergeometric representation

Set

$$\lambda = 2^{14}3^95^5, \quad \alpha = \frac{1}{30}(1, 7, 11, 13, 17, 19, 23, 29),$$

and let the seven denominator parameters be

$$\beta = \left(\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\right).$$

Cancellation of the linear factors in $A(n+1)/A(n)$ gives

$$A(n) = \lambda^n \frac{\prod_{a \in \alpha} (a)_n}{n! \prod_{b \in \beta} (b)_n}, \quad (34)$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ and $(a)_0 = 1$. One can verify the identity without invoking the gamma multiplication formula: both sides have value one at $n = 0$ and the same ratio of successive terms. It is also consistent with the hypergeometric description in A211417 [1].

The elementary identity

$$\frac{1}{kn+1} = \frac{(1/k)_n}{(1+1/k)_n}$$

shows that dividing by the three factors in (32) shifts three denominator parameters. Define

$$\beta' = \left(\frac{3}{2}, \frac{4}{3}, \frac{2}{3}, \frac{6}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\right).$$

Then

$$U_n = 42\lambda^n \frac{\prod_{a \in \alpha} (a)_n}{n! \prod_{b \in \beta'} (b)_n}, \quad (35)$$

and hence

$$G(z) := \sum_{n \geq 0} U_n z^n = 42 {}_8F_7 \left(\begin{matrix} \frac{1}{30}, \frac{7}{30}, \frac{11}{30}, \frac{13}{30}, \frac{17}{30}, \frac{19}{30}, \frac{23}{30}, \frac{29}{30} \\ \frac{2}{3}, \frac{4}{3}, \frac{2}{5}, \frac{6}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5} \end{matrix} ; \lambda z \right). \quad (36)$$

This is an equality of formal power series, and of analytic functions for $|\lambda z| < 1$. The coefficients also satisfy the convenient recurrence

$$(n+1) \prod_{b \in \beta'} (n+b) U_{n+1} = \lambda \prod_{a \in \alpha} (n+a) U_n, \quad U_0 = 42. \quad (37)$$

No recurrence is being used to define an unproved integer sequence: integrality has already been established independently.

7.3 Differential and integral transforms

Let $\theta = z \, d/dz$ and $F(z) = \sum_{n \geq 0} A(n) z^n$. Coefficient extraction gives

$$(2\theta + 1)(3\theta + 1)(5\theta + 1)G(z) = 42F(z). \quad (38)$$

Equivalently, within the common disk of convergence,

$$G(z) = 42 \int_0^1 \int_0^1 \int_0^1 F(zx^2y^3t^5) \, dx \, dy \, dt. \quad (39)$$

Termwise integration is justified by absolute convergence, and each monomial contributes the three reciprocal factors in (32).

The first-order recurrence (37) gives the differential equation

$$\left[\theta \prod_{b \in \beta'} (\theta + b - 1) - \lambda z \prod_{a \in \alpha} (\theta + a) \right] G(z) = 0. \quad (40)$$

For $n \geq 1$, its z^n coefficient is just the recurrence with index $n - 1$; its constant coefficient vanishes due to the initial factor θ . Thus G is D-finite. More generally, polynomial division of coefficients gives $P(\theta) \sum u_n z^n = C \sum R(n) z^n$. These identities make the new integer divisibility statements useful for recurrence and generating-function investigations, without needing to guess a closed algebraic equation.

8 All-order asymptotic expansions

8.1 The general factorial-ratio expansion

The usual Stirling expansion for $\log \Gamma$, including remainder control on the positive real axis, is recorded in DLMF Section 5.11 [8]. Applying it separately to the finitely many factorials in $R(n)$ yields the following explicit form. Let

$$\Lambda = \frac{\prod_i a_i^{a_i}}{\prod_j b_j^{b_j}}, \quad K_R = (2\pi)^{-h/2} \sqrt{\frac{\prod_i a_i}{\prod_j b_j}},$$

and, with B_{2q} the Bernoulli numbers, set

$$\sigma_{2q-1} = \frac{B_{2q}}{2q(2q-1)} \left(\sum_i a_i^{1-2q} - \sum_j b_j^{1-2q} \right). \quad (41)$$

Then, for any fixed $N \geq 1$,

$$\log R(n) = n \log \Lambda - \frac{h}{2} \log n + \log K_R + \sum_{q=1}^N \frac{\sigma_{2q-1}}{n^{2q-1}} + O(n^{-2N-1}). \quad (42)$$

The terms of size $n \log n$ and n cancel by balance, the $\log n$ terms give $-h/2$, and the remaining terms are exactly those displayed. Thus (42) is a Poincaré expansion to any prescribed order, not a claim that its infinite series converges.

Suppose P has degree d , positive leading coefficient ρ , and factorization (14) with all roots nonzero. For sufficiently large n , it is positive. Write

$$S_j = \sum_{i=1}^s m_i (\ell_i / k_i)^j.$$

Expanding the finitely many logarithms in $P(n) = \rho n^d \prod_i (1 + \ell_i / (k_i n))^{m_i}$ gives

$$\log P(n) = \log \rho + d \log n + \sum_{j=1}^N \frac{(-1)^{j+1} S_j}{j n^j} + O(n^{-N-1}). \quad (43)$$

Subtracting (43) from (42) proves an all-order expansion for every integer sequence constructed here. For example the first two correction terms are

$$\frac{CR(n)}{P(n)} = \frac{CK_R}{\rho} \Lambda^n n^{-d-h/2} \left[1 + \frac{\sigma_1 - S_1}{n} + \frac{(\sigma_1 - S_1)^2 + S_2}{2n^2} + O(n^{-3}) \right]. \quad (44)$$

For a negative leading coefficient, the same assertion holds with its sign restored; the logarithmic derivation is applied to $|P(n)|$.

8.2 The expansion for the certified sequence

For A one obtains

$$K_A = \frac{1}{\sqrt{60\pi}}, \quad \Lambda = \lambda, \quad \sigma_1 = \frac{7}{120}.$$

In (32), $d = 3$, $\rho = 30$,

$$S_1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{31}{30}, \quad S_2 = \frac{1}{4} + \frac{1}{9} + \frac{1}{25} = \frac{361}{900}.$$

Therefore

$$U_n = \frac{7}{5\sqrt{60\pi}} \frac{\lambda^n}{n^{7/2}} \left(1 - \frac{39}{40n} + \frac{3893}{5760n^2} + O(n^{-3}) \right). \quad (45)$$

All further coefficients are obtained by the explicit rational operations in (41)–(43). For B , the analogous base constants are $\Lambda = 2^{10}3^3$, $K_B = 1/\sqrt{12\pi}$, and $\sigma_1 = 1/36$. The method therefore gives comparably explicit expansions for every companion sequence in the sharp-constant table.

9 Verification record, limitations, and next questions

9.1 What was actually checked

The archive contains two different kinds of computational work. The finite certificates are part of the proof of the exact constants; the regression tests merely provide additional protection against implementation and transcription errors. The separate certificate checker completed successfully for all 60 graphs, all 1,685 states, and all 19,277 transition inequalities, including prime coverage and sharpness witnesses.

The supplementary tests passed 90,298 comparisons between two forms of the Landau step function, 808 direct rational checks of the eight sharp quotients, and 480,048 small-prime valuation checks over $0 \leq n \leq 10,000$. They also passed 3,336 direct checks of the explicit uniform-product bounds, 4,000 valuation checks at reproducibly sampled 256-bit indices, and 668 classification checks. The random seed was 20260919. A further 203 exact checks verify the hypergeometric recurrences and the first asymptotic coefficients. These tests are recorded in `data/test_results.json`; they are not cited as substitutes for Theorem 3.1.

Every sporadic parameter list in the atlas was separately checked for balance, height one, and nonnegativity at all rational breakpoints of its Landau function. The atlas's parameter order follows Coserea's OEIS table [7]. The derived jump-sign columns are generated from (9), not inferred from a finite list of sequence values.

9.2 Limits of the conclusions

The polynomial criterion applies to polynomials splitting over $\mathbb{Q}$. It does not classify arbitrary irreducible quadratic or higher-degree polynomials. Nor does the height-one sign rule extend unchanged to higher height: there one must use the actual capacities in (15), and multiplicities greater than one may occur. The bound (16) is uniform and constructive, but often much larger than the minimum; optimal constants for general products remain a useful computational problem.

No Lean formalization was produced or run. The general proofs are written in conventional mathematics, and the finite-graph certificates are checked by ordinary exact-integer Python code. There is no reliance on floating-point root finding, numerical asymptotics, or a conjectured recurrence in the proof of integrality. A Lean development could split naturally into the valuation identity, rational one-sided step lemmas, the prime-power cutoff argument, and a generic finite-potential checker.

9.3 Research directions arising from the proof

A natural continuation is to classify nonsplit polynomial divisors. Our necessity proof realizes prescribed rational approach points by primes in arithmetic progressions. For higher-degree factors, roots modulo primes and their distribution take the place of these simple progressions, so the rational-root proof cannot simply be copied.

A second question is the behavior, as r grows, of the *optimal* uniform multipliers for the two product families. The explicit lcm bounds settle existence but do not predict the exact small-prime exponents. The finite-state method suggests an effective route for fixed r , with a separate challenge of controlling the state space uniformly in r . Finally, the newly certified sequences invite combinatorial models that would explain their integrality without prime valuations; no such interpretation is assumed here.

A Admissible denominators for all 52 sporadic ratios

In the following table, a and b are the numerator and denominator parameter lists of $R(n)$. The column K_- contains the k with $J(k) = -1$ and therefore describes primitive factors $kn + i$, $i > 0$, $(i, k) = 1$. The column K_+ contains $J(k) = +1$ and describes $kn - i$. The signs refer to jumps, not to the sign of the offset. Every allowed root must still be simple. The parameter data are transcribed from Coserea's table [7]; the two K columns are calculations from Theorem 4.1. Row 1 is B and row 31 is A .

Index	a	b	K_-	K_+
1	12, 1	6, 4, 3	1, 2, 3	12
2	12, 3, 2	6, 6, 4, 1	1, 2, 6	12
3	12, 1	8, 3, 2	1, 2, 8	6, 12
4	12, 3	8, 6, 1	1, 2, 8	3, 12
5	12, 3	6, 5, 4	1, 2, 5	3, 12
6	12, 5	10, 4, 3	1, 2, 10	6, 12
7	18, 1	9, 6, 4	1, 2, 3, 4	18
8	9, 2	6, 4, 1	1, 2, 4, 6	9
9	9, 4	8, 3, 2	1, 2, 8	9

Continued on next page

Index	a	b	K_-	K_+
10	18, 4, 3	9, 8, 6, 2	1, 2, 8	18
11	9, 1	5, 3, 2	1, 2, 5	9
12	18, 5, 3	10, 9, 6, 1	1, 2, 10	18
13	18, 4	12, 9, 1	1, 3, 12	2, 18
14	12, 2	9, 4, 1	1, 9	2, 6, 12
15	18, 2	9, 6, 5	1, 3, 5	2, 18
16	10, 6	9, 5, 2	1, 9	2, 6, 10
17	14, 3	9, 7, 1	1, 9	2, 14
18	18, 3, 2	9, 7, 6, 1	1, 7	2, 18
19	12, 2	7, 4, 3	1, 7	2, 6, 12
20	14, 6, 4	12, 7, 3, 2	1, 3, 12	2, 14
21	14, 1	7, 5, 3	1, 3, 5	2, 14
22	10, 6, 1	7, 5, 3, 2	1, 7	2, 6, 10
23	15, 1	9, 5, 2	1, 2, 9	15
24	30, 9, 5	18, 15, 10, 1	1, 2, 18	30
25	15, 4	12, 5, 2	1, 2, 6, 12	15
26	30, 5, 4	15, 12, 10, 2	1, 2, 3, 12	30
27	15, 4	8, 6, 5	1, 2, 6, 8	15
28	30, 5, 4	15, 10, 8, 6	1, 2, 3, 8	30
29	15, 2	10, 4, 3	1, 2, 4, 10	15
30	30, 3, 2	15, 10, 6, 4	1, 2, 4, 5	30
31	30, 1	15, 10, 6	1, 2, 3, 5	30
32	15, 2	10, 6, 1	1, 2, 6, 10	15
33	15, 7	14, 5, 3	1, 2, 14	15
34	30, 5, 3	15, 10, 7, 6	1, 2, 7	30
35	30, 5, 3	15, 12, 10, 1	1, 2, 4, 12	30
36	15, 6, 1	12, 5, 3, 2	1, 2, 4, 12	15
37	15, 1	8, 5, 3	1, 2, 4, 8	15
38	30, 5, 3, 2	15, 10, 8, 6, 1	1, 2, 4, 8	30
39	20, 3	12, 10, 1	1, 2, 6, 12	20
40	20, 6, 1	12, 10, 3, 2	1, 2, 3, 12	20
41	20, 1	10, 8, 3	1, 2, 3, 8	20
42	20, 3, 2	10, 8, 6, 1	1, 2, 6, 8	20
43	20, 1	10, 7, 4	1, 2, 7	20
44	20, 7, 2	14, 10, 4, 1	1, 2, 14	20
45	20, 3	10, 9, 4	1, 2, 9	20
46	20, 9, 6	18, 10, 4, 3	1, 2, 18	20
47	24, 1	12, 8, 5	1, 2, 4, 5	24
48	24, 5, 2	12, 10, 8, 1	1, 2, 4, 10	24
49	24, 4, 1	12, 8, 7, 2	1, 2, 7	24
50	24, 7, 4	14, 12, 8, 1	1, 2, 14	24
51	24, 4, 3	12, 9, 8, 2	1, 2, 9	24
52	24, 9, 6, 4	18, 12, 8, 3, 2	1, 2, 18	24

B Reproducing the certificates and data

The implementation requires Python 3.10 or later and uses only the standard library. From the archive root, run

```
python code/verify_certificates.py
python code/test_research.py
```

The first command checks the supplied finite certificates without calling the generator. To regenerate all finite graphs, equality witnesses, sequence terms, and the atlas, run

```
python code/generate_artifacts.py
python code/verify_certificates.py
```

The checker deliberately raises an error if any state set is not closed, any transition violates its potential inequality, a required prime is missing, or a sharpness witness fails its direct valuation calculation. The large-prime assertion is supplied by the proved cutoff theorem and is not implemented as a search up to an arbitrary prime bound.

The central library also exposes `Ratio`, `Factor`, `classify`, `one_sided_capacity`, and `effective_bound`. For example:

```
from factorial_divisibility import A, Factor, classify, effective_bound

assert classify(A, (Factor(2, 1), Factor(3, 1), Factor(5, 1)))
assert not classify(A, (Factor(4, 1),))
assert not classify(A, (Factor(2, 1, multiplicity=2),))
cutoff, valid_multiplier = effective_bound(A, (Factor(2, 1),))
# The returned multiplier is a proved bound, not necessarily minimal.
```

When using the library from the archive root, add `code/` to `PYTHONPATH`, or run the example from within the code directory. Nonprimitive factors must be normalized and their common integer content treated separately. Equal roots must be merged by adding multiplicities before calling the classifier. This avoids silently mistaking two scalar multiples of a factor for two distinct roots.

The article compiles with ordinary pdfLaTeX and standard TeX Live packages. Run `make pdf`, or run `pdflatex article.tex` twice. The two atlas files and the certificate JSON are included, so compilation does not require a network connection or regeneration.

References

- [1] The OEIS Foundation, *A211417: $(30n)!n!/((15n)!(10n)!(6n)!)$* . Comments by Peter Bala dated August 28, 2025; proof-status note by Ralf Stephan dated June 30, 2026. Accessed September 19, 2026. <https://oeis.org/A211417>.
- [2] The OEIS Foundation, *A295431: $(12n)!n!/((6n)!(4n)!(3n)!)$* . Uniform-divisibility conjectures by Peter Bala dated August 28, 2025. Accessed September 19, 2026. <https://oeis.org/A295431>.
- [3] P. Bala, *Proof of the integrality of A211417 and A211418*, April 2012; revised October 2015. https://oeis.org/A211417/a211417_1.txt.
- [4] J. W. Bober, *Factorial ratios, hypergeometric series, and a family of step functions*, Journal of the London Mathematical Society (2) **79** (2009), 422–444. Preprint arXiv:0709.1977. <https://arxiv.org/abs/0709.1977>. DOI: 10.1112/jlms/jdn078.
- [5] T. Adamczewski, *OEIS Open: How many conjectures can language models turn into theorems?*, arXiv:2608.11941, August 2026, Appendix A, entry A211417. <https://arxiv.org/abs/2608.11941>.
- [6] K. S. Kedlaya, *Notes on analytic number theory*, Chapter 4, “Primes in arithmetic progressions,” Theorem 4.2 and its proof. <https://kskedlaya.org/ant/chap-primes-in-ap.html>. Accessed September 19, 2026.
- [7] G. Coserea, *Table with the parameters of the 52 sporadic integral factorial ratio sequences*, supplementary data for OEIS A295431. <https://oeis.org/A295431/a295431.txt>. Accessed September 19, 2026. Numerical parameter lists are reproduced in the accompanying data with attribution; the admissibility columns are computed in this work.
- [8] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic expansions*, especially formula 5.11.1 and the remainder discussion. <https://dlmf.nist.gov/5.11>. Accessed September 19, 2026.